

**Std : 9 Topic: Motion - Distance and Displacement**• **Learning Objectives**

- Students will be able to
- Lists the objects which are at rest and which are in motion around us
- Understand distance and displacement
- Classify the motion of vehicles as uniform motion and non-uniform motion.
- Gives examples for non uniform motion
- Distinguish between speed and velocity.

**Std : 8 Topic: Electricity and Heat- Electricity**

Students will be able to

- Acquires the knowledge of Electricity
- Understands the role of Electricity in our daily life
- Understands the three kinds of circuits
- Gives the examples of various conductors and insulators
- Distinguish between conductors and insulators.

**REVIEW QUESTIONS**

1. Write the meaning and nature and scope of Physical Science.
2. What are the aims and objectives of teaching Physical Science in Schools?
3. What are the need and significance of teaching Physical Science?
4. Write the values of teaching Physical Science?
5. Explain Cognitive Domain with its parts. Write the Instructional objectives of each part.
6. Explain Revised Blooms' Taxonomy.
7. Explain Psychomotor Domain with its parts write the instructional objectives of each part.
8. Explain Affective domain with its parts. Write the Instructional objectives of each part.

**UNIT - II****TEACHING SKILLS****2.1 Micro Teaching**

A Teacher make use of a number of methods and techniques to make the teaching effective, successful and interesting. The technique includes motivating, explaining, questioning, writing on the black board, using teaching aids and so on. The teacher could also make use of non verbal behaviours such as smiling, nodding and gesturing. These group of activities are called skills. So teaching is a various combination of skills.

Microteaching introduces to the teacher trainee to a wide range of teaching skills and allows the teacher trainee to practice each skill one at a time until he or she becomes proficient in the skill.

**2.1.1 Microteaching - Definition**

The word Microteaching was formed by A.W.Dwight Allen during 1963 defined microteaching "as a scaled down teaching encounter in class size and class time".

Allen Eve defined microteaching "as a system of controlled practice, that makes it possible to concentrate on specific teaching behaviour and to practice teaching under controlled conditions."

Microteaching is reduced teaching. Here new skills are polished - M.C. Knight 1971.

Microteaching is a teacher training technique. Here a teacher trainee teachers a short lesson of about five minutes duration to a small group of students practicing one teaching skill at a time.

### 2.1.2 Need and importance of Microteaching

In teaching learning process, we can't use single skill. At the same time if a teacher wants to be good in teaching, he should be given training in all teaching skills one by one. For this micro teaching is more useful.

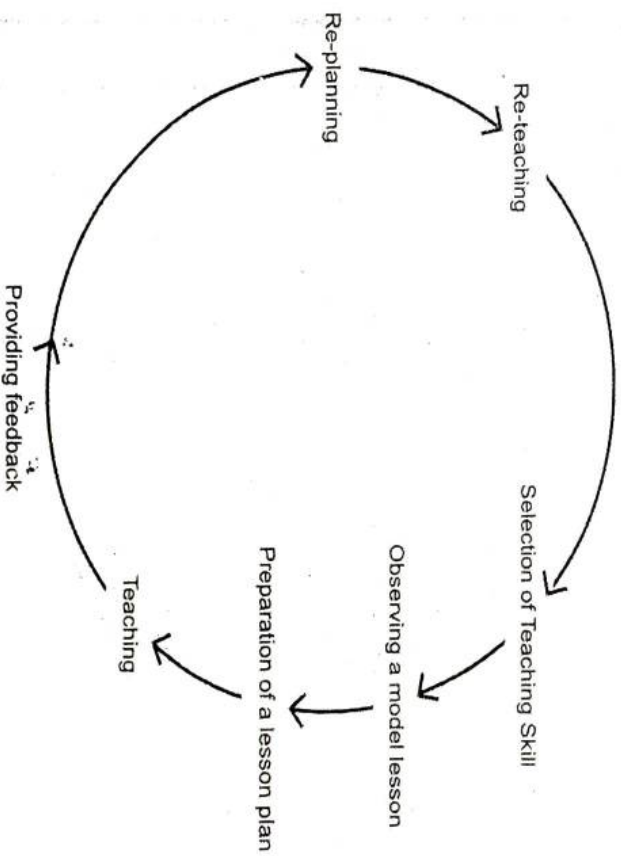
Here each skill is looked liked through microscope. Instead of "What to teach", how to teach is considered. The beauty of this method is, teacher trainee get unproblematic artificial atmosphere instead of natural class room atmosphere. This method is very helpful to improve the teaching skills and to remove the difficulties in teaching.

### 2.1.3 Characteristics of microteaching

- Microteaching is a teacher teaching technique and not a teaching method.
- It is a real teaching but if focuses on developing teaching skill and competencies.
- In microteaching the teacher trainee practices one specific teaching skill at a time, till he/she attains mastery over the skill.
- Class size is reduced to 5 - 6 students.
- Duration of the class is 5 - 6 minutes
- Microteaching is a cyclic process, the predecided model is plan, teach, feedback, re-plan, re-feedback. It is repeated till the trainee achieves the expected level of mastery.
- Microteaching is practiced in terms of definable, observable, measurable and controllable teaching skills.

### 2.1.4 Microteaching Cycle

Microteaching is a method that enables a teacher - trainee to practice one teaching skill till he or she attains mastery over the skill. The practice phase is the most important phase of the microteaching programme. In this phase the teacher trainee chooses a specific skill, prepares a micro-lesson plan and teachers a small group of students for a duration of 5 - 6 students. We follow few steps to get training in the skill. These steps are called microteaching cycle.



#### Steps in microteaching cycle

1. **Selecting a skill** : This involves selection of a teaching skill practiced, knowing its components clearly.



2. **Observing model lesson** : Teacher educator will present a model microteaching lesson plan of the selected skill. Trainee observe it carefully
3. **Preparing Lesson Plan** : Teacher trainee has to prepare a micro-lesson plan for 5 - 6 minutes for the selected teaching skill. In lesson plan all the components of the skill should be presented in an appropriate place.
4. **Teaching** : The trainee teaches the lesson for 5-6 minutes. There will be 5-6 students or peers with teacher educator or one or two peers as supervisor. If possible record the teaching methods of the trainee by video or audio tape.
5. **Feed back**: The supervisor analyses the performance and discusses it with the teacher trainee. The feedback should focus on components of the teaching skill and draw attention to modification for mastering the skill.
6. **Replanning**: In the light of the feedback received from the supervisor or peer observers, the teacher trainee replans her micro-lesson plan by writing another lesson plan or modifying the existing one on the basis of feedback.
7. **Reteaching**: The teacher-trainee reteaches the revised lesson plan to another group of students. The supervisor checks to see whether there is any improvement in skill attainment. This process repeats till the teacher trainee acquires the required level of competency.

### 2.1.5 Uses of microteaching

Microteaching techniques increases the effectiveness of the teacher training programmes in the following ways

1. Microteaching helps in reducing the complexities of the normal classroom teaching. This helps the teacher trainees gain more confidence in classroom teaching.
2. Microteaching gives the knowledge of various teaching skills.
3. It is useful for developing teaching efficiency.
4. Microteaching prepares trainee to effective and better teacher.
5. Microteaching creates among the teacher trainees an awareness of the various skills of which teaching is composed of.
6. Each trainee makes progress in developing teaching skills at his own speed depending his ability.
7. Using video or Audio tape enable the trainee to analyse his own teaching performance.
8. Feedback enable the teacher-trainee to consciously concentrate on specific behaviour modification.\

### 2.1.6 Micro Teaching Vs Macro Teaching

| Sl.No. | Micro Teaching                      | Macro Teaching                         |
|--------|-------------------------------------|----------------------------------------|
| 1.     | It is a teacher training technique. | It is a method of classroom teaching   |
| 2.     | Class size is 5 to 10 students      | Class size is more than 40             |
| 3.     | Duration of the class 6-10 minutes  | Duration of the class 40 to 45 minutes |
| 4.     | One or two teaching Skills used     | Maximum Teaching skills are used.      |

| Sl.No. | Micro Teaching                                                    | Macro Teaching                                         |
|--------|-------------------------------------------------------------------|--------------------------------------------------------|
| 5.     | Importance is given to skill                                      | Importance is given to content                         |
| 6.     | It puts the teacher under the microscope                          | No such arrangement for macro teaching                 |
| 7.     | All the faults of the teacher are observed                        | No such arrangement for macro teaching                 |
| 8.     | Feed back is given                                                | No feed back is given                                  |
| 9.     | Opportunity to reteach the lesson                                 | No such arrangement                                    |
| 10.    | No importance is given to students needs                          | Importance is given to students needs                  |
| 11.    | No classroom activities for students                              | Classroom activities will be given.                    |
| 12.    | It tends to reduce the creativity of the teachers                 | It increases the creativity of the teachers            |
| 13.    | Micro teaching can be successful in a controlled environment only | Macro teaching can be successful in any environment    |
| 14.    | Teaching is relatively simple                                     | Teaching becomes complex                               |
| 15.    | Micro teaching is done with all sensory resources                 | Macro teaching relies on auditory learning and recall. |

## 2.2 Identification of teaching skills:

A teaching skill is a set of activities or actions of the teacher which are repeated a number of times during the course of teaching.

### Characteristics Of Teaching Skills

- Teaching skills are the specific instructional activities.
- Teaching skills are verbal and non-verbal in nature.
- A teaching skill includes activities and actions.
- Teaching skills involve the teacher's activities which are performed in classroom.
- Teaching skills develop the communicative competency.
- Teaching skills help effective presentation of content in classroom.

### List of Teaching skills given by Allen and Ryan

- Stimulus variation
- Set induction
- Closure
- Teacher silence and non-verbal cues
- Reinforcing pupil participation
- Fluency in questioning
- Probing questioning
- Use of higher order questions
- Divergent questions
- Recognising and attending behaviour
- Illustrating and use of examples



13. Planned repetition
14. Completeness of communication

#### List of Teaching skills given by B.K. Passi

1. Writing instructional objectives
2. Introducing a lesson
3. Fluency in questioning
4. Probing questioning
5. Explaining
6. Illustrating with examples
7. Stimulus variation
8. Silence and non-verbal cues
9. Reinforcement
10. Increasing pupil participation
11. Using blackboard
12. Achieving closure
13. Recognizing attending behaviour

#### Types of Teaching skills selected

Large number of teaching skills have identified in bringing effective teaching. Allen and Ryan listed fourteen teaching skills and in 1983 Menon gave a list of seventy four teaching skills. A few important skills listed below will be discussed here.

1. Skill of Set induction
2. Skill of Explaining
3. Skill of questioning
4. Skill of probing questions

5. Skill of stimulus variation
6. Skill of Reinforcement
7. Skill of closure

#### 2.2.1. Skill of Introducing the Lesson / Skill of set induction

In teaching, the way of introducing a lesson is more important. The proverb "Good beginning is half of the work completed." is also fit for teaching. The teacher, before introducing a lesson, has to motivate the students and establish cognitive and affective rapport with the students. The following are the components of skill of introducing the lesson.

1. Testing previous knowledge
  - a. Preliminary attention gaining
  - b. Use of previous knowledge
  - c. Use of appropriate device
  - d. Link with a new topic
2. Attention Gaining
  - a. Use of voice to focus attention
  - b. Use of audio visual aids
  - c. Use of gesture / eye contact
  - d. Introducing something unusual
3. Motivation stimulation
  - a. To arouse curiosity
  - b. Method of introducing a lesson
  - c. Able to involve students

### 1. Testing previous knowledge

#### 1a. Preliminary attention gaining

Tell the significance of the topic. Give importance of new knowledge in our life situations.

#### 1b. Use of previous knowledge

Before giving new knowledge, teacher has to test the previous knowledge which is to be used in the new topic.

#### 1c. Use of appropriate device

By way of asking few questions or asking students to write two or three sentences about the new topic.

#### 1d. Link with a new topic

Integrate the old knowledge with the new knowledge. Use the knowledge obtain from Text book, Library book etc., to teach the new topic.

### 2. Attention Gaining

#### 2a. Use of voice to focus attention

Use clear and bold voice to focus attention.

#### 2b. Use of audio visual aids

Use appropriate teaching aids required to teach the topic to arouse pupil's attention. Charts, slides and power point presentation may be used.

#### 2c. Use of gestures / eye contact

Gestures are indication of conveying messages by body movements. Eye contact is a signal for lack of interest in a person. Much of eye contact shows that it is a threatening attitude.

#### 2d. Introducing something unusual

Relevant to the new lesson / topic, show a chart, photograph or a instrument which is not seen by student,

### 3. Motivation stimulation

#### 3a. To arouse curiosity

To bring in interest on the topic the teacher must refresh the students properly. He / She should enact to the situation. Must quote a lot of day to day happenings of the new knowledge and give a lot of examples. Self involvement is needed to bring in interest among students. He must become a part of lesson, only then he can arouse curiosity in the students.

#### 3b. Method of introducing a lesson

Special care should be given to introduce a lesson. Teacher has to select any method / technique to introduce a lesson from the methods given below according to the nature of the lesson.

1. Telling relevant statements to teach the new topic.
2. Introduce the lesson after obtaining answers from the students.
3. By way of anecdotes, Explaining or commenting, introduce a lesson.
4. By story telling technique, introduce a lesson.
5. By the method of dramatisation, introduce a lesson.
6. Using teaching aids, we introduce a lesson.
7. By the method of demonstration, introduce a lesson.

#### 3c. Able to involve students

While introducing a lesson teacher can ask students to collect or prepare materials / Aids well in advance.

### 2.2.2. Skill of Explaining

For an effective learning, explanation in teaching should be done effectively. Explaining bridges the gap in



understanding the new knowledge by relating it to the past knowledge. Thus explaining depends upon the past knowledge, the type of new knowledge and the relationship among them. Explanation can be more effective by using simple and clear language for clarity. The following are components of the skill of explanation.

1. Beginning statement
2. Explaining links
3. Concluding statement
4. Questions to test pupil's understanding
5. Irrelevant statements
6. Lacking in continuity
7. Vague words and phrases
1. **Beginning statement:**

While explaining the beginning statement must include central concept of the content. Statements such as 'Today we are going to drive a formula or prove a theorem' etc. Beginning statement must gain attention of the students.

## 2. **Explaining links:**

While explaining the subject matter use suitable link words between the concepts. examples: therefore, that is why, since, because, hence, in order to etc.

## 3. **Concluding statement:**

Concluding statement should summarise what the teacher explained. Students can summarise what they learn if concluding statements are meaningful.

## 4. **Questions to test pupil's understanding**

Questions asked here are from the content explained. Teacher check the understanding of students by questioning. These questions will also help to analyse the performance of the explanation given by the teacher.

## 5. **Irrelevant Statement**

Teachers should avoid irrelevant statements while explaining the topic.

## 6. **Lacking in continuity**

Knowledge given in classroom are information, facts, concepts etc. It should be given in sequence and in continuity. Then only the students clearly understands the concepts, facts and information.

## 7. **Vague words and phrases**

The teacher should not use vague words and phrases which is not understandable by students. The vocabulary used should be clear and according to the level of students.

## 2.2.3 Skill of questioning

Successful teaching technique is highly depend on questioning. It is a skill that can be used in teaching any subject and any standard. In a person different types of questions, depending upon situation and purpose to be achieved are used. Questions are used to help students recall certain facts, exercise their reasoning ability, and use their power of recognition and discrimination in enhancing their learning. Effective questioning encourages them to reflect over things and ideas and leads them to participate in discussion.

## **Purpose of questioning**

- Questioning is a logical procedure of problem solving

- Questioning is useful for presenting the tentative nature of knowledge.
- The teacher encourages the learners to seek more than one answer for a question
- The teacher uses questioning to achieve learning objectives
- The teacher uses questioning to assess students understanding.

Thus questioning promotes involvement, enhance learning and motivate students.

Let us examine why questioning is an essential skill for a teacher.

- It helps students test their knowledge
- It explicit desired information from students
- It stimulates thinking
- It develops subject matter
- It enable students to analyse the problem
- It enable students to apply their knowledge to a specific new situation.
- It helps students evaluate for themselves their understanding of the subject matter.
- It motivate students to participate in the teaching-learning process.

Questioning stimulates thinking. The level of students thinking depends upon the level of questions asked by the teacher. The level of question is also determined by the level of the responses given by the students. If order to elicit responses at different levels from the students, we can frame questions at different levels.

Questioning is one of the most important skill in teaching. It has to be developed by any teacher. While teacher teaches a lesson he has to ask several questions related to the content. These type of questions, are called teaching questions. After completing a lesson, questions will be asked to know whether the students understood the content of teaching. These questions are called review questions. Teachers should be expert in asking questions. Good questions serve many purposes. Questions stimulate thinking. Questions help recalling facts, Concepts and principles. Questions make active involvement in learning. Questions arouse the curiosity of the students

Generally these are three levels of question. They are 1. Low order questions 2. middle level questions 3. higher order questions.

#### 1. Low order questions.

These questions require the students to recall knowledge that they have learnt. The questions of this level do not ensure understanding of the content or analyzing information, but requires memorization.

Examples :

1. What are the various types of mixtures?
2. Give reason for air is a mixture
2. **Intermediate level of questions.**

Explaining the meaning of a concept or statement, compare or differentiate among two things etc are intermediate level of questions.

1. What is the difference between mixture and compound?
2. Explain how the speed of a vehicle is measured by a traffic control vehicle?



### 3. Higher order questions

These type of questions requires the students to use higher levels of intellectual abilities such as comprehension, application, analysis, synthesis and evaluation.

**Example:**

1. Sound waves cannot pass through vacuum or empty space. Describe an experiment to illustrate the above statement.
2. With the help of simple pendulum prove that  $1/T^2$  is constant.

### 2.2.4 Skill of probing

The skill of probing questions involves going deep into pupil's response by asking a series of subsequent questions in a topic. It helps the pupils to give a correct and complete answer. This skill requires the teacher to put a series of questions about the answer given to the first question. It is a major micro-teaching skill.

When students are unable to give the answer or give incomplete answers, teachers use probing questions.

#### **Purpose of probing**

1. To motivate the students for correct answer
2. Ask many questions related to the main subject
3. Taking various supportive questions
4. Linking previous knowledge with the new knowledge
5. Creating interest and curiosity among students.
6. Developing the thinking, creativity and analytical ability

#### **Components of probing skill of questioning**

There are five main components of skill of probing

### Teaching Skills

1. Prompting
2. Seeking further information
3. Re-Focusing
4. Re-Direction
5. Critical awareness

#### **1. Prompting**

Prompting mostly involves verbal questions, visual questions and some visual guidance. In order to help the student to know the correct answer, the teacher gives some indication.

#### **2. Seeking further information**

Most of the time students don't give the proper and accurate answer. Then the teacher asks the same question to other students in order to get the correct answer. The teacher asks different aspects of the question in order to seek further information. This also may check the students' knowledge.

#### **3. Re-Focusing**

If the teacher is not satisfied with the answer of the student then the teacher puts different situations in front of the students which helps them in the transfer of knowledge of the students.

#### **4. Re-Direction**

The teacher tries to know the multiple answer to the question in order to make students think again and again. This will increase the thinking ability of the students, and this may increase the participation of the students in the class too.

## 5. Critical awareness

If the teacher asks why? how? type of questions then it is called Critical Awareness. This critical awareness among students increase the rational thinking and awareness.

### 2.2.5. Skill of stimulus variation

While teaching teacher must draw attention of the students. For this, teacher uses some gestures, body movements, makes certain verbal statements, modulating the voice etc. All these behaviours are related to stimulus variation. Components of this skill are :

1. Movements
2. Gestures
3. Change in speech pattern
4. Change in inter action style
5. Secure attention
6. Pausing
7. Oral - Visual switching
1. **Movements**

Teacher should not stand in one place while teaching. He has to move from one place to another to write on the blackboard, to go near and appreciate a student, to move towards a model etc. Hence, in order to gain students attention a teacher has to move purposeful movements.

## 2. Gestures

Use of appropriate gestures according to the content to emphasis importance, to explain emotions, make students to understand the message. Hand movement to describe shape, size, Body movements and facial expression enhance the value of message.

## 3. Change in speech pattern.

If a teacher explains the content in the same modulation, then it is boring. Hence, in teaching there should be voice modulation to pay attention to a particular point. Modulating the voice on situations, change in pitch, speed of voice will force his student to pay attention.

### 4. Change in interaction style

Teacher has to plan, how students also involve in teaching. Asking questions from the students, asking student to do a sum on the blackboard, students are encouraged to put few questions and teacher will answer it, to participate students in a seminar or discussion and teacher will tell its result etc. are all called change in interaction. This can be mention in a simple form as

Teacher - Class

Teacher - Student

Student - Student

### 5. Secure Attention.

While teaching the teacher may use certain statements like, this is important, look at this diagram, listen carefully etc to draw attention. While writing on the blackboard teacher draw attention by using colour chalk for important message or underlined the important points.

### 6. Pausing

Pausing means introducing silence during teaching. From this act, we can draw attention of the students. While teaching continuously or at the time of asking questions teacher has to plan pausing and create attention.

### 7. Oral - Visual switching

While the teacher orally explains, student's hearing. When the teacher exhibits aid or writes on blackboard his



visual sense is used. Such frequent changes from oral to visual or from visual to oral help to secure pupil's attention.

### 2.2.6 Skill of Reinforcement / Skill of Appreciation

All these behaviours are related to stimulus variation. The skill of reinforcement refers to the effective use of reinforcers. While teaching, a teacher asks questions and students replies it with correct answer. If the teacher simply goes to another question, then students will not happy. If the teacher reinforce a correct answer with a smile or praise the student, then they will be happy. This is called reinforcement. Teacher has to practice using positive reinforcers and avoid the use of negative reinforcers in the classroom. By Reinforcement, students participation is more in classroom activities. The different components of this skill are

1. Positive verbal reinforcement
2. Positive non verbal reinforcement
3. Negative verbal statement
4. Negative non verbal reinforcement
5. Wrong use of reinforcement
6. Inappropriate use of reinforcement

#### 1. Positive verbal reinforcement

When the students give correct response for a question the teacher has to give positive verbal reinforcement such as Good, Very good, Well, Excellent, Continue etc.

#### 2. Positive non verbal reinforcement

When a student give correct response for a question the teacher may give non-verbal reinforcement expressing happiness or with the help of body language. Teachers

should use more positive reinforcers so that the students participation is maximum in teaching activities.

#### 3. Negative verbal statement

Teacher should avoid using these type of negative verbal reinforcers namely, I do not like you, stop it, nonsense, you won't improve, you will fail in the examination etc.

#### 4. Negative non verbal reinforcement

Teacher should not use these type of Non-verbal negative reinforcers such as facial expression like, frowning, sadness, moving away from responding student, looking at angrily etc.

#### 5. Wrong use of reinforcement

When a student gives a wrong answer, teachers should avoid using words like idiot, stop, enough or showing terror face etc. which will hurt student and his enthusiasm will decrease in the subject.

#### 6. Inappropriate use of reinforcement

Teacher should not use reinforcers in a delayed manner or using it always. Using the above type of reinforcement affects the learning and decrease the motivation of the students.

### 2.2.7. Skill of closure

After teaching a subject matter in a better way, it is better to close it in the same way. Then only the students feel pleasure and satisfaction. At the end of the lesson, it is necessary to make a quick review. What has been taught, applying new knowledge in various situations, associating new knowledge with old knowledge and repeating important points. Various components of this skill are:

1. Consolidation of major points.
2. Application of present knowledge.



3. Linking post knowledge with present knowledge.
4. Linking present knowledge with future learning.

### 1. Consolidation of major points

While teaching teacher has explained so many facts to explain the subject matter. At the end he has to summarise it in a meaningful way. Then only students identify what they have learned.

### 2. Application of present knowledge

After teaching new knowledge, the teacher has to explain how and where it is used with examples. Then only the students know the importance of new knowledge and its application. For this the teacher may use problem solving method.

### 3. Linking past knowledge with present knowledge

After explaining the new knowledge, teacher should link it with the past knowledge. Then only the students get a clear view about what they learned and feel that their knowledge has developed.

### 4. Linking present knowledge with future learning

The new knowledge learned by the students should be linked to the knowledge what they will go to study in future or in continuation to this knowledge. From this they will be ready to learn more about this knowledge.

### 2.2.8 Advantages or merits of Micro teaching

- It offers opportunity to practice a real lesson without the complexities of a normal classroom.
- It offers several sources of immediate feedback.
- One skill is learnt at a time which is observable, controllable and practicable.
- It helps in the development of self-confidence.



- It helps the trainees to develop the skill of blackboard writing.
- It is helpful to improve his/her own teaching.
- It makes trainees familiar with various teaching styles eg. Black-board writing, use of aids, asking questions etc.
- Evaluation procedure of a micro-lesson is more precise objective and acceptable.
- It saves time and energy of the student-teacher as well as the teacher - educator.
- It focusses attention on the modification and improvement of teacher behaviours in the teaching learning process.

### 2.3 Model Micro Lessons

#### Model Micro Lesson – 1

|                               |                             |
|-------------------------------|-----------------------------|
| Name of the Student Teacher : | xxxxx                       |
| Skill :                       | Skill of Stimulus variation |
| Standard :                    | IX B                        |
| Subject :                     | Science - Physics           |
| Topic :                       | Is matter around us pure?   |
| Duration :                    | 5 Minutes                   |
| Date :                        | xxxxx                       |

#### Components of skill

1. Movement



2. Gestures
3. Change in speech pattern
4. Change in interaction style
5. Focusing
6. Pausing
7. Oral visual switching

### Episode

| Teacher / student | Content                                                                                                           | Components of the skill |
|-------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------|
| Teacher           | Good Morning Students                                                                                             |                         |
| Student           | Good Morning Madam                                                                                                |                         |
| Teacher           | Today we will see the matter surrounding us in this class. Teacher goes near to a student and ask what is matter? | Movement                |
| Student           | All objects having mass & volume is called matter                                                                 |                         |
| Teacher           | Teacher shook his head appreciating the answer In how many forms Matter exists?                                   | Gestures                |
| Student           | Matter exists in three forms                                                                                      |                         |
| Teacher           | Very good. What are they? Write in on the blackboard?                                                             |                         |
| Student           | Writes the three forms of the matter on the blackboard                                                            | Change in interaction   |

| Teacher / student | Content                                                                                                                                     | Components of the skill           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Teacher           | Teacher once again told that the matter exists in three forms namely solid, liquid and gas with a pause and using change in speech pattern. | Pause & Change in Speech pattern. |
| Teacher           | Whether the molecules are held tightly together in definite arrangements in solids                                                          |                                   |
| Student           | Yes. The molecules are held tightly together in definite arrangements                                                                       | Change in interaction style       |
| Teacher           | Your answer is correct. Teacher showed a piece of chalk piece as an example.                                                                | Oral visual switching             |

### Model Micro Lesson – 2

Name of the Student Teacher : xxxxx

Skill : Skill of reinforcement

Standard : VIII<sup>th</sup>

Subject : Science

Topic : Coal and Petroleum

Duration : 5 Minutes

Date : xxxxx

### Components of the skill

1. Positive verbal reinforcement



2. Positive non verbal reinforcement
3. Negative verbal statement
4. Negative non verbal reinforcement
5. Wrong use of reinforcement
6. Inappropriate use of reinforcement

### Episode

| Teacher / student | Content                                                 | Components of the skill           |
|-------------------|---------------------------------------------------------|-----------------------------------|
| Teacher           | Good Morning Students                                   |                                   |
| Student           | Good Morning Sir                                        |                                   |
| Teacher           | What are the modes of transport?                        |                                   |
| Student           | Bus, Car, Scooter, Cycle etc.                           |                                   |
| Teacher           | Very Good. Ram, tell how Bus, Car or Scooter is moving? | Positive verbal reinforcement     |
| Ram               | With the help of Diesel or Petrol Sir.                  |                                   |
| Teacher           | Teacher shook his head appreciating the answer          | Positive non verbal reinforcement |
| Teacher           | What is the fuel used in cooking?                       |                                   |
| Student           | LPG Sir                                                 |                                   |
| Teacher           | Good. What is the expansion of LPG                      |                                   |
| Student           | Light Petrol Gas Sir                                    |                                   |

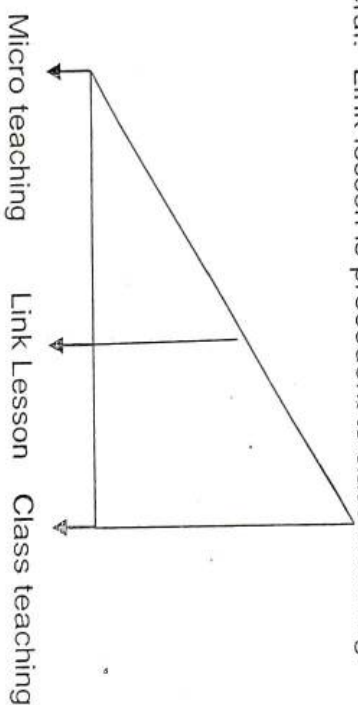


| Teacher / student | Content                                                                                                                                                                                                                       | Components of the skill       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Teacher           | It is a simple answering you don't know. Teacher looked the student with anger                                                                                                                                                | Negative non verbal statement |
| Teacher           | Sankar, you tell what is the expansion of                                                                                                                                                                                     |                               |
| Student           | It is Liquid fied Petroleum Gas                                                                                                                                                                                               |                               |
| Teacher           | Nodding his head                                                                                                                                                                                                              | Positive non verbal statement |
| Teacher           | Let us see what are fuels. Substances that born in air to give heat and energy are called fuels.                                                                                                                              |                               |
|                   | Fossil fuels are formed from the buried remains of decayed plants and animals over millions of years under the influence of heat and pressure in the absence of air. Coal, Petroleum and natural gas are called fossil fuels. |                               |
| Teacher           | Ram will you tell the names of few fossil fuels.                                                                                                                                                                              |                               |
| Ram               | Coal, Petrol and Natural gas                                                                                                                                                                                                  |                               |
| Teacher           | Good                                                                                                                                                                                                                          | Positive Verbal statement     |



## 2.4 Link Lesson

In micro teaching each teaching skill is taken separately and training is given in that skill. All these skills are combined in class teaching. In micro teaching importance is given to teaching skill then the content. In Link Lesson a small content and three or four teaching skills are taken. Importance is given for both content and teaching skills in link lesson. Here teacher trainee is teaching 10 - 15 students in 10 - 15 minutes. In this way if we give few teaching classes to the teacher-trainee to take link lessons and then if we send to class teaching, then it will be more useful. Link lesson is precedent to class teaching.



Micro teaching → Link Lesson → Class teaching

### Understanding major Steps in teaching a Link Lesson

While preparing a Link lesson plan, the following steps are followed :

#### Designing a Link Lesson

The following steps are used in designing a Link Lesson plan.

1. Selecting a content / topic.
2. Setting Lesson objectives.
3. Resource materials needed.
4. Content outline

## Teaching Skills

5. Teaching skills used
6. Instructional Procedures

### 1. Selecting a Content / topic

The first thing to consider obviously is what you want to teach. This should be based a unit / topic of your syllabus. You also need to be aware of what standard you are developing a topic plan and also the time given for your teaching.

### 2. Setting Lesson objectives

To make sure your topic plan will teach exactly what you want it to do, you need to develop clear and specific objectives. Note that objectives are the learning outcomes of the lesson. You can certainly have more than one objective for a topic plan.

### 3. Resource materials needed

Find out what resource materials you are giving to use to explain the topic plan and prepare it well in advance. Be sure the teacher will have materials everything to teach the topic. Power point presentation, Flash Cards, charts or materials are very helpful in teaching your topic.

### 4. Content outline

Write content key points to be presented.

### 5. Teaching Skills used

Write the name of the teaching skills used. While teaching a link lesson.

### 6. Instructional Procedures

- a. Motivation
- b. Presentation

- c. Interaction
- d. Reflection
- e. Summing up

#### a. Motivation

Students are encouraged and motivated to learn new knowledge. The mind of the child should be prepared to receive new knowledge. Preparation may involve.

1. Testing previous knowledge to learn new knowledge.
2. Integrating the previous knowledge with the new knowledge.
3. Arrest attention of the students using new techniques.
4. Announcing the aim of the lesson.

#### b. Presentation

Here teaching takes important place. Students learn new knowledge. For an effective learning outcome, teacher should ensure active student participation with the use of audio-visual aids. Teacher should stimulate thinking by asking number of questions while teaching.

#### c. Interaction

Interaction means the student teacher activities while teaching is going on in the class. Interaction leads learning. All interaction will take place to attain the objectives.

#### d. Reflection

A process of self-observation and self-evaluation is called reflection. By collecting information about what goes on in our classroom and by analysing and evaluating this information is called reflection. We identify and explore our

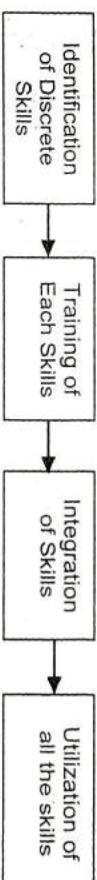
success and failures. This may then lead to changes and improvements in teaching. Reflection can be given by self, peer people or students. Reflection can also be seen with the help of audio and video recording of class teaching.

#### e. Summing up

After teaching a subject matter in a better way, it is better to close it in the same way. Then only the students feel pleasure and satisfaction. At the end of the lesson, it is necessary to make a quick review what has been taught, and repeating important points.

#### Practising a Link Lesson

#### Concept of Link teaching



#### Link Lesson plan

A Link lesson is a short lesson that provides instruction in one or more skills or concept that students will relate to a larger lesson plan that follow. Link lessons can be used to teach particular skills, create interest in a topic and generate questions or introduce strategies.

The art of teaching, writing, the Link lesson allows a teacher to convey a content or strategy to students that they will use often. It allows students to gain valuable relevant skills on a regular basis without spending too much of time to teach the same skills. The Link lesson should be short and its duration will be 15 to 20 minutes. Here the teacher introduce the topic, demonstrate skills or concept; guide student practice; discuss the topic; give more examples; and task about what was taught. Link lessons can be taught individually, in a small group or to a whole class.



**Link lesson plan format**

Name of the Teacher Trainee :

Date :

Standard and section :

Subject : Physical Science

Unit :

Topic :

Instructional objectives :

- 1.
- 2.
- 3.

**Instructional Resources Required**

- 1.
- 2.

**Content outline****Teaching Skills used****Instructional procedures****Link Lesson - Lesson Plan**

Name of the Teacher Trainee : xxxxx

Date : xxxxx

Standard and Section : VII Std. C

Subject : Science

**Teaching Skills**

Unit : Matter in our surroundings

Topic : Properties of matter

Time : 20 minutes

**Instructional objectives**

1. Student acquires knowledge of matter.
2. Student understand what is matter.
3. Understands the characteristics of the molecules of matter

**Instructional Resources Required**

Beacher, Water, Sugar, Blue ink, pencil, milk, balloon, chalk piece, chalk piece powder

**Teaching skills used**

1. Skill of introducing the lesson
2. Skill of explaining
3. Skill of questioning
4. Skill of stimulus variation
5. Skill of non-verbal cues
6. Skill of reinforcement
7. Skill of closure

**Content outline**

Motivation – Characteristics of matter – Characteristics of molecules – Three states of matter – Examples – Explaining with experiments – recapitulation.

| Content/<br>Concept                                                                 | Specificati<br>on of<br>behavior-<br>ural<br>objectives     | Learning<br>Experience<br>(Teacher/<br>Learner<br>activities)                                                                                                                                       | Evaluation                                                                                            |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Introductory<br>activities<br>Motivation                                            | Recalls                                                     | Teacher shows pencil water, balloon with air, book etc and ask few questions on the state of matter.                                                                                                | Give any two examples of solid.<br>Give any two examples of liquid.<br>Give any two examples of gases |
| Development<br>activities<br>Introducing the<br>topic<br>properties<br>of<br>matter | Acquires<br>knowledge                                       | Teacher explains the difference between matter and molecules with the help of chalk piece and chalk powder.<br>Atoms and molecules are very small. We can measure this with the unit of Nano metre. | Molecular<br>area<br>We can<br>measure<br>atoms with<br>the unit of                                   |
| Explaining with<br>the help of a<br>experiment                                      | Character-<br>istics of<br>molecules<br>of<br>the<br>matter | Teacher Explains them is space in between molecules with a experiment                                                                                                                               | There is<br>between<br>molecules                                                                      |

| Content/<br>Concept                                                                                                                     | Specificati<br>on of<br>behavior-<br>ural<br>objectives | Learning<br>Experience<br>(Teacher/<br>Learner<br>activities)                             | Evaluation                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Interaction<br>Add<br>Foundain Pen<br>ink slowly in the<br>beaker and keep<br>it some time<br>without any<br>movement and<br>observe it | Explains<br>a<br>experiment                             | Teacher explains that molecules continuously moving and mix each other with an experiment | Molecules<br>moves and<br>_____                                               |
| Things found in<br>solid, liquid and<br>gaseous state                                                                                   | State of<br>things<br>(Analysis)                        | Teacher explains the three states of things to the student clearly                        | Give few<br>examples of<br>solid                                              |
| Reflection                                                                                                                              |                                                         | Students gave the correct responses for all the questions                                 | 1. Give any two examples of solid.<br>2. What is the unit to measure an atom? |
| Concluding<br>activities                                                                                                                | Recapitu-<br>lation                                     | At the end Teacher recall all the important points of matter.                             |                                                                               |

**Link Lesson - Episode**

Skill : Integration of skills  
Class : VII Std. A  
Subject : Science



Unit : Matter is our surroundings  
Topic : Properties of matter

| Teacher / Student            | Teaching – Learning Activities                                                                                   | Skill Components                                    |
|------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Motivation</b><br>Teacher | Good Morning Students.                                                                                           |                                                     |
| Students                     | Good Morning Sir.                                                                                                |                                                     |
| Teacher                      | Teacher shows Pencil, Chalk, book, milk, water, balloon with air etc. and ask to give any to examples of solids. | Skill of introducing<br>Skill of questioning        |
| Students                     | Pencil, book                                                                                                     |                                                     |
| Teacher                      | Good. Give any two examples of liquid state things.                                                              | Skill of reinforcement, Skill of questioning        |
| Students                     | Water, Petrol                                                                                                    |                                                     |
| Teacher                      | Very good.                                                                                                       | Skill of reinforcement                              |
| Teacher                      | Give any two examples of gaseous state                                                                           | Skill of questioning                                |
| Students                     | Water vapour, Oxygen                                                                                             |                                                     |
| Teacher                      | Good. Very good. Nodding his head and appreciating the answer                                                    | Skill of reinforcement,<br>Skill of non-verbal cues |
| <b>Presentation</b>          | Today we will see what is a matter and know its                                                                  | Skill of explaining                                 |

## Teaching Skills

| Teacher / Student            | Teaching – Learning Activities                                                                                                                                                                                                                                                                                                                       | Skill Components                                    |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Teacher                      | Physical Characteristics. Things which have mass and value is called matter.                                                                                                                                                                                                                                                                         |                                                     |
| Teacher                      | Teacher calls a student and give him a piece of chalk and ask him to make it powder.<br>These particles of chalk powder make chalk piece and is called matter. Small particles are made of molecules. Molecules are made of atoms. Atoms and molecules are very small We measure those with the unit of Nano meter (1 Nano meter = $10^{-9}$ metres) | Skill of stimulus variation, Skill of explaining    |
| Teacher                      | Teacher wrote "The unit to measure the atom is _____" on the black board and asked Sankar to complete the fill in blank.                                                                                                                                                                                                                             | Skill of questioning<br>Skill of stimulus variation |
| <b>Interaction</b><br>Sankar | Sankar filled the blank with correct answer                                                                                                                                                                                                                                                                                                          |                                                     |
| Teacher                      | Good                                                                                                                                                                                                                                                                                                                                                 | Skill of reinforcement                              |

| Teacher / Student            | Teaching – Learning Activities                                                                                                                                                                                               | Skill Components                                 |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Teacher                      | Teacher take a beaker with water and adds sugar in it and mixed thoroughly. Kandasamy where is the sugar?                                                                                                                    | Skill of explaining                              |
| Kandasamy                    | It dissolves in water                                                                                                                                                                                                        |                                                  |
| Teacher                      | Good. Whether water level in the beaker increased?                                                                                                                                                                           | Skill of reinforcement                           |
| Kandasamy                    | No Sir.                                                                                                                                                                                                                      |                                                  |
| Teacher                      | When sugar dissolves in water. It filled the gap of water molecules. Hence the water level of the beaker has not increased from this experiment. We know there is space between molecules                                    | Skill of explaining                              |
| Teacher                      | Again teacher take a beaker with water and slowly add fountain pen ink and ask student to observe it. From this experiment we see ink slowly mingles with water. We see molecules continuously moves and mingles each other. | Skill of stimulus variation, Skill of explaining |
| <b>Summing up</b><br>Teacher | Let us see the important points discussed here.                                                                                                                                                                              | Skill of closure.                                |

| Teacher / Student | Teaching – Learning Activities                                                                                                             | Skill Components |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|                   | 1. What is the unit to measure the molecules?<br>2. There is _____ between molecules of liquid.<br>3. What are the three states of matter? |                  |

### Difference between microteaching, Link lesson and class teaching

| Characteristics | Micro teaching     | Link lesson         | Class teaching           |
|-----------------|--------------------|---------------------|--------------------------|
| Time            | 5 - 6 minutes      | 15 - 20 minutes     | 40 - 45 minutes          |
| No. of students | 5 - 6              | 10 - 15             | 40 - 50                  |
| Content         | Very Small content | Small content       | One Lesson or part of it |
| Teaching skills | Only one skill     | Three or four skill | Various skills           |